

An A/B test is an experiment:

- All elements are constant
- Except for one variable

2 variants:

- A/B tests

More than 2 variants:

- A/B/N tests

- A/B/N tests
- Randomized controlled experiments
- Controlled experiments
- Split tests

Goal:

Make data driven decisions

When results are **reliable** and **repeatable**

Make results reproducible:

Change to product → change to metrics

“

Randomized controlled experiments are
the gold standard for establishing causality.

”

“

We believe online controlled experiments are:

- the **best scientific way** to establish causality with high probability

”

- A/B/N tests
- Randomized controlled experiments
- Controlled experiments
- Split tests

“

We believe online controlled experiments are:

- able to **detect small changes** that are hard to detect with other techniques, such as change over time.

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EXPERIMENT PREREQUISITES

1. Define key metrics

- Overall Evaluation Criterion (OEC)

2. Changes are easy to make

- Difficult changes introduce complexity
- Eg. Re-designing the entire website
may be too difficult.

3. Have enough "randomization units"

- Randomization unit:

"Who" or "what" randomly allocated to different groups

- How many is enough?

Thousands

Larger number → detect smaller effect

EXPERIMENT DESIGN

1. What population to select

- Specific population vs. all users

EXPERIMENT DESIGN

1. What population to select

- Specific population vs. all users

2. Size of an experiment

- Statistical power → sample size

2. How long to run an experiment

- Seasonality
- Day of week
- Primacy and novelty effect

RUNNING EXPERIMENTS

Collect data

- Instrument loggings
- Utilize companies platform

RESULT TO DECISION

Check & interpret results to make decisions

- Data Scientists spend the most time

When making decisions, consider

- Tradeoffs between different metrics

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User engagement 

Revenue 

When making decisions, consider

- Cost of launching a change

Sanity checks

- Passed → continue with analysis
- Failed → discard the results, look into root cause

When making decisions, consider

- Cost of launching a change

Eg. Engineering maintenance

Opportunity cost

When costs are high:

- Benefit should outweigh the cost
- Set practical significance boundary

When costs are low:

- Launch any positive changes

Long-term effect

- Different from short-term effect